Portfolio Optimization and Market Risk Analysis: Evidence from Berkshire Hathaway, AT&T, and U.S. Equities

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1. Executive Summary

This study consists of two analyses. The primary analysis focuses on Berkshire Hathaway (BRK-B) and AT&T (T), utilising the Markowitz model to determine the efficient portfolios. The secondary analysis focuses on the market correlation of thirty U.S. stocks by computing their Beta via the Single Index Model.

Sections 2 to 4 provide a comprehensive overview of the study's contents. Section 2 provides readers with general information about the two chosen companies by discussing their background and business descriptions. Section 3 explained two relevant theoretical frameworks: the Markowitz Model and the Single Index Model. The Markowitz model is used to perform risk-return analysis and determine the minimum-variance portfolio and the optimal portfolio. The single index model is used to determine a stock's beta by regressing an individual stock's return against the market's return. Section 4 outlines the data sources and the Excel analysis methods applied in the study.

The main findings suggest that BRK-B has a lower return and lower risk than T during the sampling period, consistent with the high-risk, high-return theory. Empirical results suggest a moderate positive correlation ($\rho = 0.4613$) between them, indicating inferior portfolio diversification. The findings support that the minimum variance portfolio is achieved by overweighting BRK-B (75.92%) and underweighting T (24.08%) in the portfolio. In contrast, the optimal risky portfolio requires underweighting BRK-B (24.48%) and overweighting T (75.52%). Regarding the Beta computation, seventeen stocks had negative Beta, while thirteen stocks had positive Beta. However, only one stock has a Beta absolute value greater than one, and only three stocks' betas are statistically significant at a 95% confidence level.

2. Company Background and Business Description

2.1. Berkshire Hathaway (BRK-B)

Berkshire Hathaway originally began as a textile manufacturing company in the 19th century. However, its transformation into the company we recognise today began in the 1960s, when Warren Buffett took control through his investment partnerships (TIOmarkets, 2024). Buffett transformed Berkshire Hathaway from a struggling textile firm into a diversified holding company. Under his leadership, the company pursued a long-term value investing approach. It began by buying entire businesses as well as taking significant minority stakes in publicly traded companies (Berkshire-Hathaway Co., n.d.).

Berkshire Hathaway is now a highly diversified conglomerate investing across a wide range of industries and securities. Its core business areas include insurance and reinsurance (with companies like GEICO and General Re), freight rail transport (through BNSF Railway), and a combination of manufacturing, service, and retail operations (InsuranceBusiness, 2022; Samuel, 2023). It also holds significant equity positions in leading publicly traded companies, such as Apple, Coca-Cola, and American Express. Its investment portfolio spans both domestic and foreign markets, which subjects it to global economic trends and the industry life cycles (Stone, 2025).

In recent years, Berkshire Hathaway has further strengthened its market position through huge investments and aggressive stock repurchases. It has also expanded its role in the tech sector, especially with its significant stake in Apple (Stone, 2025). Berkshire Hathaway maintains a highly competitive position in the market, thanks to its substantial cash holdings, which enable a flexible business approach and maintain a steady focus on long-term goals. Its sophisticated investment decision-making, combined with the leadership of investment legend Warren Buffett and his wingman, Charlie Munger, as well as the newly appointed CEO, Greg Abel, has made it a trusted name in global finance (Sharma, 2025).

2.2. AT&T (T)

AT&T, the American Telephone and Telegraph Company, was founded in 1885 and emerged from the Bell Telephone Company. Alexander Graham Bell, the inventor of the telephone, initially founded it. AT&T was formed to operate the original long-distance telephone system and become a major player in U.S. telecommunications (Gianforti, 2022). This monopoly, known as the Bell System, was authorised in 1913 by the Kingsbury Commitment. It was a state monopoly for much of the 20th century until its dissolution in 1984, and created a group of regional companies called the "Baby Bells." The modern AT&T Inc. was formed in 2005 when SBC Communications, one of the Baby Bells, acquired AT&T Corp. and changed its name, marking a new era of expansion and innovation (Wikiwand, 2024).

AT&T is a multinational telecommunications company that provides a range of services, including Wireless (AT&T Communications), wireline (which includes AT&T Fibre), and business solutions for enterprise customers. The Wireless segment offers mobile and broadband services to individuals and companies, while the Wireline segment provides high-speed internet and multimedia solutions, such as AT&T Fibre, to millions of households. AT&T Business provides enterprise solutions, including cloud, cybersecurity, and networking

(CSIMarket.com, 2024). The corporation also operates FirstNet, a dedicated network for public safety departments. Furthermore, AT&T is a leader in 5G deployment and fibre internet expansion in the U.S., serving over 140 million wireless subscribers and maintaining a significant presence in both urban and rural markets (Pier, 2025). Additionally, the company competes with Verizon, T-Mobile, and Comcast in the telecom sector, while its media ventures face rivals such as Netflix and Disney (Taylor, 2025).

AT&T has prioritised streamlining operations and enhancing its primary connectivity services in recent years. In 2022, the company sold its media holdings, including WarnerMedia, merging it with Discovery Inc. to form Warner Bros. Discovery, and focusing on telecommunications and network innovation (Warner Bros. Discovery, 2022). It has made significant investments in expanding its 5G and fibre networks, targeting fibre availability to over 30 million sites by 2025. AT&T remains one of the world's top three telecoms companies in terms of market capitalisation and brand value. Despite severe competition from Verizon and T-Mobile in the wireless industry, the company has a competitive advantage due to its large scale, extensive infrastructure investments, and a strategy focused on long-term growth and customer experience (Narcisi, 2022).

3. Theoretical Framework

3.1. Markowitz Model

The Markowitz Model, also known as Modern Portfolio Theory (MPT), was introduced by Harry Markowitz in 1952 (Team, 2025). It is a useful investment strategy for risk-averse investors to minimise risk for a given level of return, or to maximise the expected return of a portfolio for a given level of risk. The model enables investors to perform a risk-return analysis by identifying the minimum-variance portfolio and the optimal tangency portfolio, thereby achieving their objectives.

The minimum-variance portfolio is the portfolio composed of risky assets on the efficient frontier that offers the lowest possible risk, measured by variance (Gajendrakar, 2025). This portfolio is particularly beneficial for risk-averse investors, as it focuses purely on minimising risk, without necessarily maximising returns. In contrast, the optimal tangency portfolio is the best possible portfolio on the efficient frontier that lies tangent to the Capital Market Line (CML). It delivers the highest Sharpe ratio, which indicates the best possible risk-adjusted return. In fact, the Sharpe ratio represents the slope of the CML, reflecting the risk premium per unit of risk.

Before calculating the risk and return of a portfolio, it is essential to consider how the assets are combined, as each combination involves different calculation methods. To determine the minimum-variance portfolio and optimal tangency portfolio, calculations of the asset weights, expected portfolio return, portfolio variance, standard deviation, and coefficient of variation are required. Then, the asset combination with the lowest variance is identified as the minimum-variance portfolio. At the same time, the asset combination with the highest Sharpe ratio is recognised as the optimal tangency portfolio.

However, the Markowitz model assumes that investors are rational and risk-averse, which is not realistic. The model also relies on the assumption of a strong and efficient market, which does not hold in reality. Furthermore, the model relies heavily on historical data to forecast future returns, which may result in inaccurate predictions (*Challenging Modern Portfolio Theory*, 2021). Despite its limitations, understanding the construction and significance of the minimum-variance and optimal tangency portfolios is essential in this study, as it demonstrates how investors can apply the Markowitz Model to achieve efficient diversification and make informed investment decisions based on their risk preferences.

3.2. Single Index Model

The Single Index Model (SIM) is a technical analysis tool that is widely used in financial analysis to assess the risk and return characteristics for individual securities within a portfolio. This model is based on the core assumption that the returns of a security can be explained mainly by a market index that broadly represents the overall market performance. For example, investors may use the S&P 500 index or the CSI 300 index as a market benchmark to predict the expected performance of an individual stock by observing the relationship between the index's return and the stock's return.

The Single Index Model assumes the total return of a security is composed with two components which are systematic risk and unsystematic risk. Systematic risk is caused by the fluctuations in the overall market that are non-diversifiable across all assets. In contrast, unsystematic risk is the unique risk associated with a particular security that can be diversified away. The Single Index Model uses linear regression to describe the mathematical link between the market return and the returns of the security to quantify this relationship (Team, 2025b).

$$Ri = \alpha i + \beta iRm + \varepsilon i$$

The Single Index Model formula shows that the R_i represents the expected return of a security, R_m is the return of the market return, α_i is the excess return of the security, indicating the additional return generated by the security after controlling for market factors, β_i is the beta measuring the security's sensitivity relative to the market, and Ɛ represents the error term, or the unsystematic risk (Fajasy, 2025).

Beta value is the primary indicator for assessing the volatility of a security. If the volatility is higher than that of the overall market, the risk and return exhibit greater fluctuation, and the beta value is greater than 1. Conversely, a lower beta value measures that the security is relatively stable and less sensitive to market fluctuations. This result the less the asset is affected by the overall market condition (Kenton, 2025).

SIM is typically used in financial fields, such as asset pricing, portfolio construction, and risk management, due to its simplicity and practicality. Investors can utilize the SIM model to assess the fair value of securities and gain a deeper understanding of the systematic and unsystematic risk structure of the entire portfolio, thereby enabling more informed investment decisions (Team, 2025b). SIM provides a framework that allows investors to identify sources of risk, predict potential returns, and effectively allocate assets to achieve investment goals by systematically analysing the returns of securities. The Single Index Model holds a significant position in financial analysis and is an indispensable component of modern investment theory.

4. Data and Methodology

4.1. Data Collection

The stock and market data used in the analysis are sourced from Investing.com. The data comprises daily records spanning from January 2 to April 1, 2025, with a total of 61 closing prices for the stocks and index. The closing prices are used to calculate each day’s holding period. The 61-day closing price is used to compute a 60-day holding period return, and our analysis is based on these 60 data points.

4.2. Analysis Methods

The holding period return (HPR) is computed using Equation 1, where Pt-1 represents the stock price at time t-1 and Pt represents the stock price at time t. Both stocks are assumed to have no dividend payments during the period. The average and the standard deviation of the HPR of two stocks are calculated using the Excel functions. The coefficient of variation is calculated using Equation 2, which illustrates the degree of volatility or risk assumed in exchange for the anticipated return on investments (Hayes, 2025). The covariance and correlation coefficient of the two stocks are calculated based on Equations 3 and 4.

$$HPR = \frac{P_{t-1} - P_t + Income}{P_t}$$

Equation 1: HPR Formula (How to Calculate the Holding Period Returns, 2012)

$$CV\ (\%) = \left(\frac{Standard\ deviation}{Mean}\right) \times 100$$

Equation 2: Coefficient of Variation Formula (Bradburn, 2019)

$$\mathrm{Cov}_{ij} = \sum_{i=1}^{n} [R_i - E(R_i)][R_j - E(R_j)]/n$$

Equation 3: Covariance formula

$$\rho_{XY} = \frac{Covariance(\boldsymbol{X}, \boldsymbol{Y})}{\sigma_X \sigma_Y}$$

Equation 4: Correlation coefficient formula

In this study, we applied Markowitz's model for portfolio management to determine the most efficient portfolio. The weights were arranged from 0% to 100% for each asset to determine the minimum variance portfolio, as per Equation 5. Besides, the optimal risky portfolio is defined by Equation 6. The Sharpe ratio of the optimal risky portfolio, which indicates the most favourable balance of risk and return, is measured using Equation 7 (Fernando, 2025).

$$W_1 = \frac{\sigma_2^2 - \rho_{1,2}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{1,2}\sigma_1\sigma_2}$$

$$W_2 = 1 - W_1$$

Equation 5: Minimum variance portfolio weight allocation formula

$$W_A = \frac{\left(E(R_A) - R_f\right)\sigma_B^2 - \left(E(R_B) - R_f\right)\sigma_{AB}}{\left[E(R_A) - R_f\right]\sigma_B^2 + \left[E(R_B) - R_f\right]\sigma_A^2 - \left[E(R_A) - R_f + E(R_B) - R_f\right]\sigma_{AB}}$$

$$W_B = (1 - W_A)$$

Equation 6: Optimal risky portfolio weight allocation formula

$$S = \left(\frac{R_p - R_f}{\sigma_p}\right)$$

Equation 7: Sharpe Ratio Formula (Beauchamp, 2019)

For the stock Beta computation, we utilise the data analysis software packages in Excel to perform simple linear regressions. The stock's HPR are input as the Y variable, and the S&P 500 index returns are input as the X variable. The outcomes are in the form of an ANOVA, where the coefficient of the X variable is the Beta for the stock. We categorised the Betas according to their significance at a 95% level using the Excel conditional formatting function.

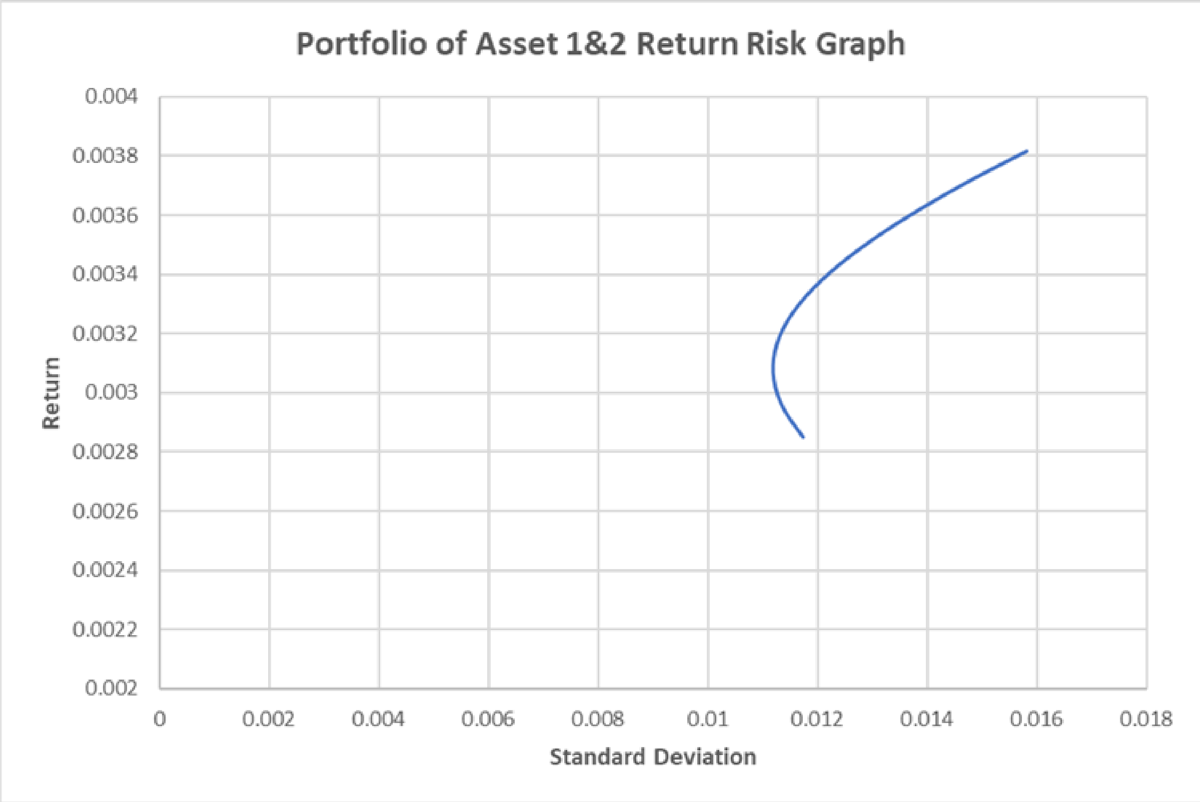
5. Main Findings

The findings, as shown in Table 5.1, were obtained during the sampling period. The average returns for BRK-B and T are 0.285% and 0.382%, respectively. Besides, the standard deviations are 0.011747 and 0.015825, respectively. The coefficient of variations are 4.11940 and 4.1473, respectively. The results align with the general theory that higher-risk investments yield higher returns. BRK-B, as a lower-volatility stock, is generating a lower return than a higher-volatility stock, T. On a risk-adjusted basis, BRK-B outperforms T due to its lower coefficient of variation. This indicates that BRK-B is taking a lower risk for each unit of return it generates. Moreover, the covariance between the stocks is 8.5761E-05, supporting that the co-movement in the stocks' prices is weak. In contrast, the correlation coefficient between them is 0.46134. This suggests that both stocks have a moderate positive relationship, and the portfolio has suboptimal diversification. Consequently, the portfolio could be better diversified by selecting lower-correlated or even negatively correlated stocks.

	Berkshire Hathaway (BRK-B)	AT&T (T)
Average	0.00285	0.00382
Standard Deviations	0.01175	0.01583
Coefficient of Variation	4.11940	4.14731
Covariance	8.5761E-05	
Correlation Coefficient	0.46134	

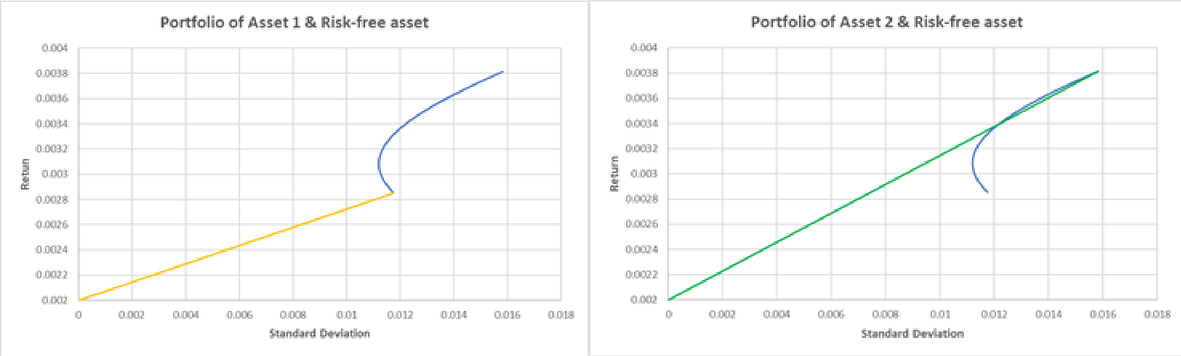
Table 5.1: BRK-B and T return-risk analysis and correlation results.

The risk-return visualisation of different weightage combinations of BRK-B and T is displayed in Graph 5.2. Some information is shown in the efficient frontier in the graph. First and foremost, the portfolio return increases gradually as more weight is allocated to T. Additionally, the efficient frontier illustrates that the lower-risk portfolio combination occurs when the investor overweights BRK-B and underweights T in the portfolio. Moreover, the concave efficient frontier indicates that there is a diminishing return as investors are taking more risk by overweighing stock T. Therefore, the optimal risky portfolio occurs at a point where stock T is overweight but not dominating the portfolio.



Graph 5.2: Portfolio of BRK-B and T with different weightage combinations.

For analysis purposes, we assume a risk-free rate of 0.2% during the sampling period to derive the Capital Allocation Line, which consists of a risky asset and a risk-free asset. The CAL reflecting portfolio of BRK-B and the risk-free asset is displayed in the left panel of Graph 5.3 (yellow line), while the CAL representing the portfolio of T and the risk-free asset is shown in the right panel (green line). The yellow CAL connects to the bottom part of the efficient frontier, whereas the green CAL connects to the top part of the efficient frontier. The underlying concept is that the portfolio return begins at 0.2% with zero risk, where all weightage is allocated to risk-free assets, and the investor earns only the risk-free return. As an investor allocates more weight to the risky asset, the portfolio risk increases, and the return also increases. The endpoint is where all weightage is assigned to the risky assets. The CAL reaches one of the edges of the efficient frontiers that analyse the risk-return tradeoff of the two risky assets.



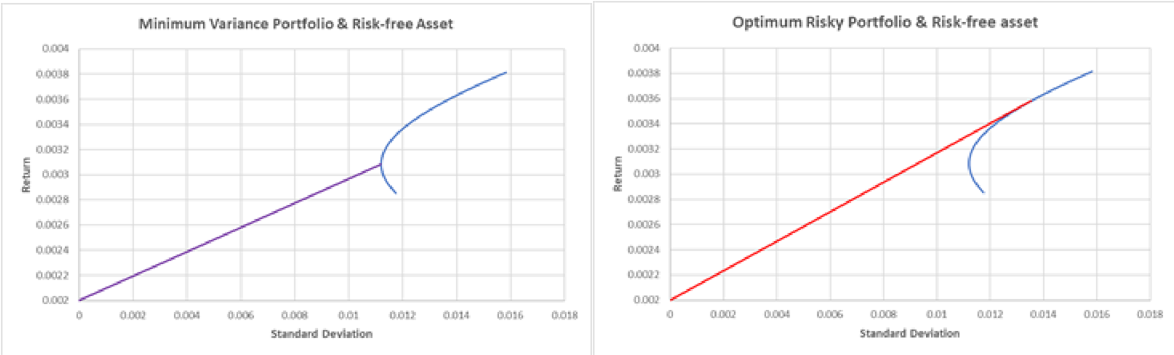
Graph 5.3: Capital Allocation Line for a portfolio of a risky asset and a risk-free asset.

The minimum variance portfolio and the optimal risky portfolio are listed in Table 5.4. The results indicated that the minimum variance portfolio should allocate 75.92% of its weight to BRK-B and 24.08% to T. This allocation will generate the lowest possible variance and, therefore, a standard deviation of 0.01120. Furthermore, the optimal risky portfolio occurs when 24.48% of the portfolio is allocated to BRK-B and 75.52% is allocated to T. This combination will attain the steepest CAL with a slope of 0.11684, where the marginal return of the additional risk reached its maximum.

	Weight of BRK-B	Weight of T	Standard Deviation / Slope
Minimum Variance Portfolio	0.7592	0.2408	0.01120
Optimal Risky Portfolio	0.2448	0.7552	0.11684

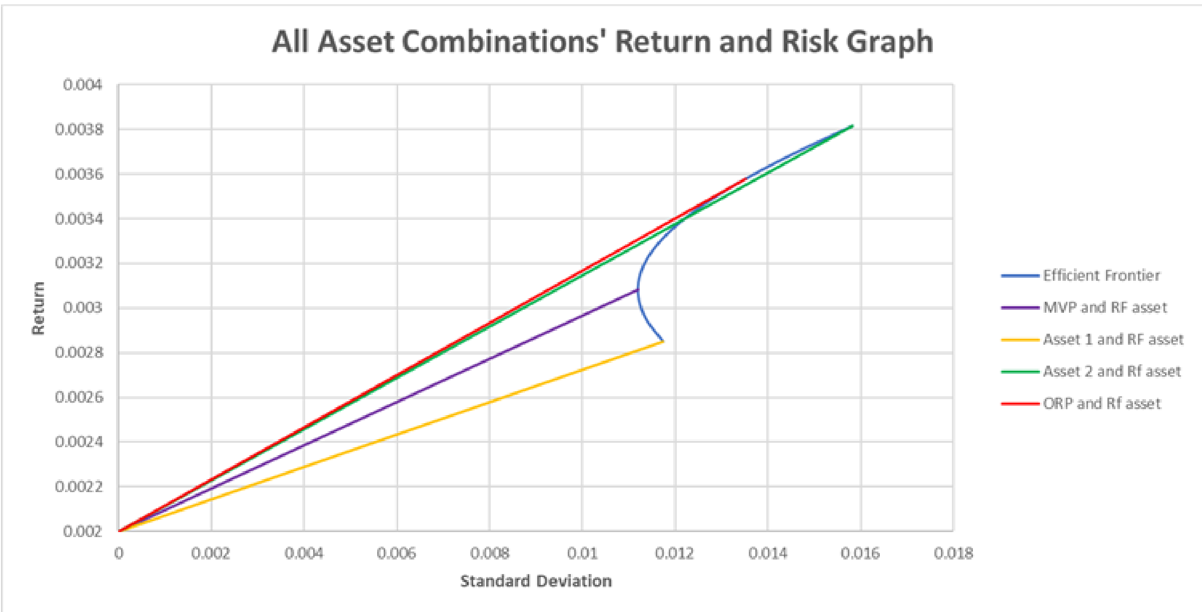
Table 5.4: The weightage allocation to attain minimum variance and optimal risky portfolio.

The CAL for the combination of the minimum variance portfolio and the risk-free asset is exhibited in the left panel of Graph 5.5. The CAL starts at a 0.2% return on the y-axis, with zero risk, and attaches to the leftmost point of the efficient frontier. The visualisation is theoretically sound, as the extreme point of the concave efficient frontier corresponds to the minimum point of the standard deviation. Besides, the CAL for the combination of the optimal risky portfolio and the risk-free asset is shown in the right panel of Graph 5.5. The CAL begins from 0.2% on the y-axis and is tangent to the efficient frontier at its steepest point. This is the point at which the Sharpe ratio is maximised and the most efficient risk-return tradeoff is achieved. This CAL is also known as the Capital Market Line.



Graph 5.5: Capital Allocation Line for minimum variance portfolio and optimal risky portfolio combining with risk-free asset.

To obtain a comprehensive view, we plot the four CALs on a single graph, as shown in Graph 5.6. According to the Markowitz Model, risk-averse investors should invest in a portfolio that places them on the Capital Market Line to achieve the optimal risk-return tradeoff. On the other hand, those investors seeking the lowest risk, regardless of the return, should invest in the position reflected in the minimum variance portfolio and risk-free asset CAL on the efficient frontier. Those seeking to maximise return without considering risk should invest in the stock T and the risk-free asset CAL. The BRK-B and the risk-free asset CAL is suggested to be inferior in terms of risk-adjusted return due to its lowest slope.



Graph 5.6: The CALs and the efficient frontier

Switching to the secondary analysis, the Betas of the 30 stocks are shown in Table 5.7. Seventeen stocks are found to be negatively correlated with the market index represented by the S&P 500 index, while thirteen stocks are positively correlated. Additionally, most stocks are insensitive to systematic risk due to a lower absolute value of Beta, with only ADBE's Beta exceeding 1 in absolute value. However, the results do not stand statistically, as only the betas of three stocks (ADBE, DIS, and DPZ) are statistically significant. This may be due to the limited time horizon of our analysis which covers only 61 days.

No	**Stocks**	**Beta**
1	Adobe Systems Incorporated (ADBE)	1.1427
2	Alibaba Group Holdings Ltd ADR (BABA)	-0.3693
3	Amazon (AMZN)	0.3443
4	American Express (AXP)	0.0304
5	Applied Materials Inc (AMAT)	-0.2886
6	ASML Holding NV ADR (ASML)	-0.0509
7	AT&T (T)	-0.1558
8	Berkshire Hathaway (BRK-B)	0.0297
9	Boeing Co (BA)	-0.2039
10	Chevron (CVX)	-0.0886
11	Coca-Cola (KO)	0.0254
12	Constellation Brands Inc Class A (STZ)	-0.2724
13	Disney (DIS)	0.3898
14	Domino's Pizza Inc (DPZ)	-0.5799
15	Goldman Sachs (GS)	0.1210
16	JD.com Inc Adr (JD)	-0.6462
17	JPMorgan (JPM)	-0.2378

18	Kraft Heinz Co (KHC)	-0.0744
19	Kroger (KR)	0.0733
20	Lockheed Martin Corporation (LMT)	0.0230
21	McDonald's Corporation (MCD)	0.2338
22	Mondelez International Inc (MDLZ)	-0.0998
23	Morgan Stanley (MS)	0.2444
24	Occidental Petroleum Corporation (OXY)	-0.1604
25	Palantir (PLTR)	-0.6638
26	Procter & Gamble Company (PG)	-0.2018
27	Salesforce (CRM)	0.3085
28	UnitedHealth Group (UNH)	-0.1213
29	Visa (V)	-0.0442
30	Walmart (WMT)	0.1720

Table 5.7: 30 stocks' Betas

6. Conclusion

In summary, this study analysed the possible portfolio combinations that consist of Berkshire Hathaway and AT&T's stock, assuming a risk-free return of 0.2% during the 61-day period. Throughout our findings, we discovered that BRK-B, as an investment holding company, has a lower risk and return profile compared to the telecommunication provider T. However, BRK-B outperforms T in terms of risk-return analysis measured by the coefficient of variation. They are discovered to have a moderate positive relationship.

As we apply the Markowitz Model, the minimum variance requires 75.92% of the weight to be allocated to BRK-B and 24.08% to T. In contrast, the optimal risky portfolio is at the point where 24.48% of the weight is assigned to BRK-B and 75.52% is assigned to T. The results indicate that investors aiming to minimise risk should invest in the minimum variance portfolio.

In contrast, those seeking the best risk-return tradeoff should invest in the optimal risky portfolio. The remaining portfolio combinations are suboptimal.

As for the secondary analysis, the 30 stocks' Betas are computed by running a regression of the stocks' return on the S&P 500 index return. The results indicate that the majority of stocks are negatively correlated with market performance, and this relationship is relatively weak. However, the results lack reliability as most of the Betas are statistically insignificant.

7. References

Beauchamp, C. (2019, January 17). The Sharpe ratio explained. *IG.* https://www.ig.com/sg/trading-strategies/the-sharpe-ratio-explained-190117

Berkshire-Hathaway Co. (n.d.). Textiles History. https://history.textiles.ncsu.edu/textile-companies/berkshire-hathaway-co/

Bradburn, S., PhD. (2019, February 19). *How to calculate the Coefficient of Variation (CV)*. Top Tip Bio. https://toptipbio.com/calculate-coefficient-variation-cv/#google_vignette

Challenging modern portfolio theory. (2021, October 14). Portfolio for the Future | CAIA. https://caia.org/blog/2021/10/14/challenging-modern-portfolio-theory-

Chen, J. (2024, January 16). *What is a Capital Allocation Line (CAL) line? How to calculate*. Investopedia. https://www.investopedia.com/terms/c/cal.asp

Csimarket.com. (2024)At And T Inc (T) Quarterly and Annual Segment Results by Country and Region - CSIMarket. (2024). Csimarket.com. https://csimarket.com/stocks/segments.php?code=T

Fajasy. (2025, April 10). *How to Calculate and Interpret the Single Index Model (SIM) | StableBread*. StableBread. https://stablebread.com/single-index-model/

Fernando, J. (2025, May 16). *Sharpe Ratio: Definition, Formula, and examples*. Investopedia. https://www.investopedia.com/terms/s/sharperatio.asp

Gajendrakar, P. (2025, May 14). Minimum Variance portfolio. *Meaning, Formula, Calculation*. https://www.wallstreetmojo.com/minimum-variance-portfolio/

Gianforti, A. (2022, January 24). *The Breakup of the Bell System*. History Factory. https://www.historyfactory.com/insights/this-month-in-business-history-bell-system/

Hayes, A. (2025, May 22). *Coefficient of Variation: definition and how to Use it*. Investopedia. https://www.investopedia.com/terms/c/coefficientofvariation.asp#:~:text=The%20coefficient%20of%20variation%20shows,of%20return%20expected%20from%20investments.

How to calculate the holding period returns. (2012, October 8). https://financetrain.com/how-to-calculate-the-holding-period-returns

InsuranceBusiness. (2022, March 11). Berkshire Hathaway : Everything you need to know. Insurance Business. https://www.insurancebusinessmag.com/asia/companies/berkshire-hathaway/398304/

Samuel, N. (2023, November 28). *Warren Buffett's bet on Railroads: Long-term Investment case study*. Picture Perfect Portfolios. https://pictureperfectportfolios.com/warren-buffetts-bet-on-railroads-long-term-investment-case-study/

Kenton, W. (2025, May 31). *What beta means for investors*. Investopedia. https://www.investopedia.com/terms/b/beta.asp#:~:text=Beta%20is%20an%20indicator%20of%20the%20price%20volatility,higher%20the%20beta%20number%2C%20the%20higher%20the%20risk.

Narcisi, G. (2022, April 5). AT&T Dumps Time Warner Business Four Years After $85B Deal. CRN. https://www.crn.com/news/networking/at-t-dumps-time-warner-business-four-years-after-85b-deal

Pier, S. (2025, April 24). AT&T gains 324,000 new subscribers in the face of Verizon losing 289,000. PhoneArena. https://www.phonearena.com/news/att-gains-subcribers-in-q1-2025_id169748

Sharma, R. (2025, June 9). *Warren Buffett's AI legacy: Apple and Amazon now make up a fifth of Berkshire's $282 billion portfolio*. The Economic Times; Economic Times. https://economictimes.indiatimes.com/markets/stocks/news/warren-buffetts-ai-legacy-apple-and-amazon-now-make-up-a-fifth-of-berkshires-282-billion-portfolio/articleshow/121729268.cms

Stone, N. (2025, April 15). *Berkshire Hathaway's Strategic Shifts: Navigating 2025 with Cash, Control, and Defensives*. Ainvest. https://www.ainvest.com/news/berkshire-hathaway-strategic-shifts-navigating-2025-cash-control-defensives-2504/

Stone, N. (2025, April 15). *Berkshire Hathaway's Strategic Shifts: Navigating 2025 with Cash, Control, and Defensives*. Ainvest. https://www.ainvest.com/news/berkshire-hathaway-strategic-shifts-navigating-2025-cash-control-defensives-2504/

Taylor, P. (2025, May 31). AT&T - statistics & facts [Review of AT&T - statistics & facts]. Statista. https://www.statista.com/topics/1252/atundt/#topicOverview

Team, I. (2025, May 17). *Modern Portfolio Theory: What MPT is and how Investors use it*. Investopedia. https://www.investopedia.com/terms/m/modernportfoliotheory.asp

Team, W. (2025b, May 12). Single Index Model. *What It Is, Formula, Assumptions, Portfolio*. https://www.wallstreetmojo.com/single-index-model/

TIOmarkets. (2024, July 1). Berkshire Hathaway: Explained | TIOmarkets. TIOmarkets. https://tiomarkets.com/article/berkshire-hathaway

Warner Bros. Discovery. (2022, April 8). Discovery and AT&T Close WarnerMedia Transaction - Warner Bros. Discovery. Warner Bros. Discovery. https://www.wbd.com/discovery-and-att-close-warnermedia-transaction

Wikiwand.com. (2024). *History of AT&T - Wikiwand*. Wikiwand.com. https://www.wikiwand.com/en/articles/History_of_AT&T